ASSIGNMENT 1

Data Exploration and Preprocessing

The Basic Exploratory Data Analysis (EDA) : As we can see in Figure 1.1, the California Housing Dataset contains information about the characteristics related to housing prices in California. This dataset is mainly used for regression problems, especially predicting housing prices based on various attributes. It consists of 9 columns and 20,640 rows.

Independent Variables (Features)

The independent variables in this dataset represent the various attributes of houses and their locations. The features are as follows:

1. **MedInc (Median Income):** The median income in a given block group (in 10,000$).
2. **HouseAge:** The median age of houses in a given block group in years.
3. **AveRooms:** The average number of rooms per household in a given block group.
4. **AveOccup:** The average number of residents per household in a given block group.
5. **Latitude:** The geographic latitude of a block group.
6. **Longitude:** The geographic longitude of a block group.

These features are used to understand the California housing market and predict housing prices. Importantly, there are no missing values in this dataset, as shown in Figure 1.2, ensuring that the analysis and modeling processes are not affected by gaps in the data.

Dependent Variable (Target)

The dependent variable is the outcome we want to predict based on the independent variables:

- **MedHouseVal (Median House Value):** This is the target variable representing the median value of houses in a given block group (in $100,000). This variable is continuous and is often used to determine the state of the real estate market and predict future trends.

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422
5	4.0368	52.0	4.761658	1.103627	413.0	2.139896	37.85	-122.25	2.697
6	3.6591	52.0	4.931907	0.951362	1094.0	2.128405	37.84	-122.25	2.992
7	3.1200	52.0	4.797527	1.061824	1157.0	1.788253	37.84	-122.25	2.414
8	2.0804	42.0	4.294118	1.117647	1206.0	2.026891	37.84	-122.26	2.267
9	3.6912	52.0	4.970588	0.990196	1551.0	2.172269	37.84	-122.25	2.611

Figure 1.1: The first 10 rows of the dataset

```
MedInc          0
HouseAge        0
AveRooms        0
AveBedrms       0
Population      0
AveOccup        0
Latitude        0
Longitude       0
MedHouseVal     0
dtype: int64
```

Figure 1.2: Check missing values

Figure 1.3 presents the basic statistics (mean, median, and standard deviation) for the numerical columns.

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	3.070655	35.631861	-119.569704	2.068558
median	3.534800	29.000000	5.229129	1.048780	1166.000000	2.818116	34.260000	-118.490000	1.797000
std	1.899822	12.585558	2.474173	0.473911	1132.462122	10.386050	2.135952	2.003532	1.153956

Figure 1.3: Statistical Summary of Numerical Columns

Data Exploration and Preprocessing

Preprocess the Data : In Figure 1.4, a sample of the dataset is shown, focusing on the preprocessing steps, including the standardization of numerical features and the creation of a binary classification target based on median house values

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal	HouseValueClass
0	2.344766	0.982143	0.628559	-0.153758	-0.974429	-0.049597	1.052548	-1.327835	4.526	expensive
1	2.332238	-0.607019	0.327041	-0.263336	0.861439	-0.092512	1.043185	-1.322844	3.585	expensive
2	1.782699	1.856182	1.155620	-0.049016	-0.820777	-0.025843	1.038503	-1.332827	3.521	expensive
3	0.932968	1.856182	0.156966	-0.049833	-0.766028	-0.050329	1.038503	-1.337818	3.413	expensive
4	-0.012881	1.856182	0.344711	-0.032906	-0.759847	-0.085616	1.038503	-1.337818	3.422	expensive

Figure 1.4: Sample of Preprocessed Data

Logistic Regression for Classification

This section details the application of logistic regression for classifying houses as 'expensive' or 'not expensive.' The dataset is divided into training and testing sets, with 80% allocated for training and 20% for testing. A logistic regression model is trained on the training set, followed by predictions on the test set.

Model performance is evaluated using several metrics, including the confusion matrix, as presented in Figure 2.1. The results indicate the following cases:

- **True Positives (TP)**: 1691 (When the actual price is expensive and the model predicts it as expensive)
- **False Positives (FP)**: 359 (When the actual price is not expensive and the model predicts it as expensive)
- **True Negatives (TN)**: 1718 (When the actual price is not expensive and the model predicts it as not expensive)
- **False Negatives (FN)**: 360 (When the actual price is expensive and the model predicts it as not expensive)

Figure 2.1: Confusion Matrix for Logistic Regression Model

Performance Evaluations:

- **Accuracy Score**: 0.83, This indicates that 83% of the predictions made by the model are correct.
- **ROC-AUC**: 0.91, This value suggests a high level of discrimination between the positive (expensive) and negative (not expensive) classes. A score above 0.90 indicates excellent model performance, meaning it can effectively distinguish between the two classes.

Positive Class Metrics (Expensive)

For the positive class, which represents expensive houses, the metrics are as follows:

- **Precision**: 0.82, This means that 82% of the houses predicted as expensive are indeed expensive, reflecting the model's ability to minimize false positives.
- **Recall**: 0.82, This indicates that the model correctly identifies 82% of all actual expensive houses, highlighting its effectiveness in capturing true positives and minimizing false negatives.
- **F1-Score**: 0.82, The F1-Score, as the harmonic mean of precision and recall, provides a balanced measure of the model's overall performance in predicting expensive houses. This reinforces the model's overall capability in accurately identifying expensive houses.

ROC Curve: Figure 2.2 presents the ROC curve, which illustrates the trade-off between the true positive rate (sensitivity) and the false positive rate across various threshold settings. The area under the curve (AUC) is a key metric that indicates the model's ability to distinguish between the positive class (expensive houses) and the negative class (not expensive houses). A higher AUC value signifies better model performance.

Overall, the model shows strong performance in predicting expensive houses, with high accuracy, precision, recall, and an excellent AUC score, indicating its ability to effectively distinguish classes and make accurate predictions.

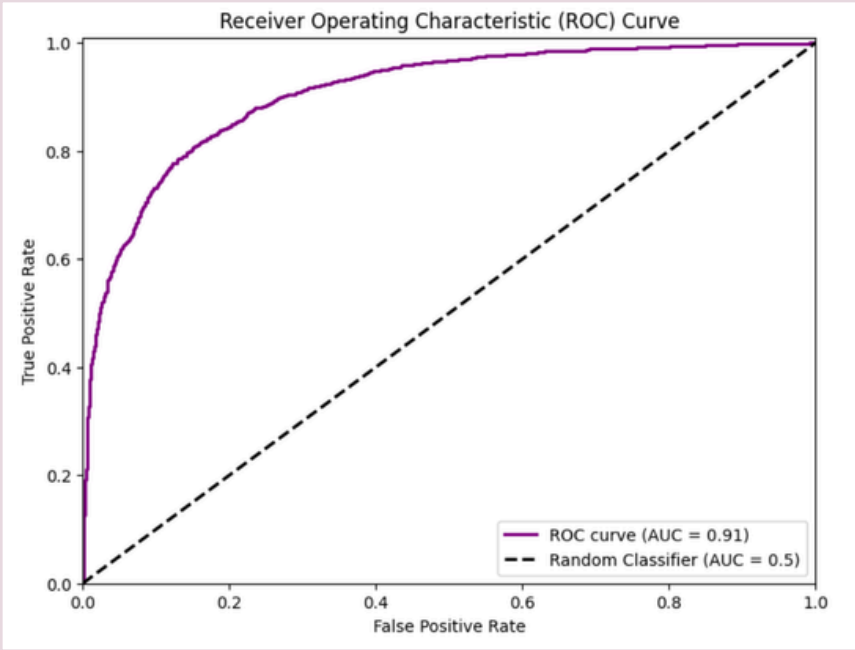

Figure 2.2: ROC Curve

Non-liner Regression

In this section, a polynomial regression model of degree 2 was implemented to predict house prices using the original continuous target variable. The dataset was divided into training and testing sets, with 80% allocated for training and 20% for testing. After training the model, predictions were made on the test set.

The performance of the non-linear regression model was evaluated using the following metrics, which indicate the model's predictive accuracy:

- **Mean Squared Error (MSE):** An MSE of 0.46 is considered good, indicating a relatively low average squared difference between predicted and actual values, which suggests that the model's predictions are close to the actual values.
- **R-squared Score:** An R^2 value of 0.65 indicates that the model explains 65% of the variance in house prices, leaving 35% unexplained, suggesting that while it captures some important patterns, other influential factors may not be included.

Overall, while the low Mean Squared Error (MSE) indicates good model accuracy, the R-squared score suggests that the model's performance is moderate. The unexplained 35% of the variance in house prices highlights the need for improvements to better capture data complexities.

Additionally, a plot comparing the actual versus predicted values for the polynomial regression model is shown in Figure 3.1, where a straight line represents the ideal scenario of perfect predictions.

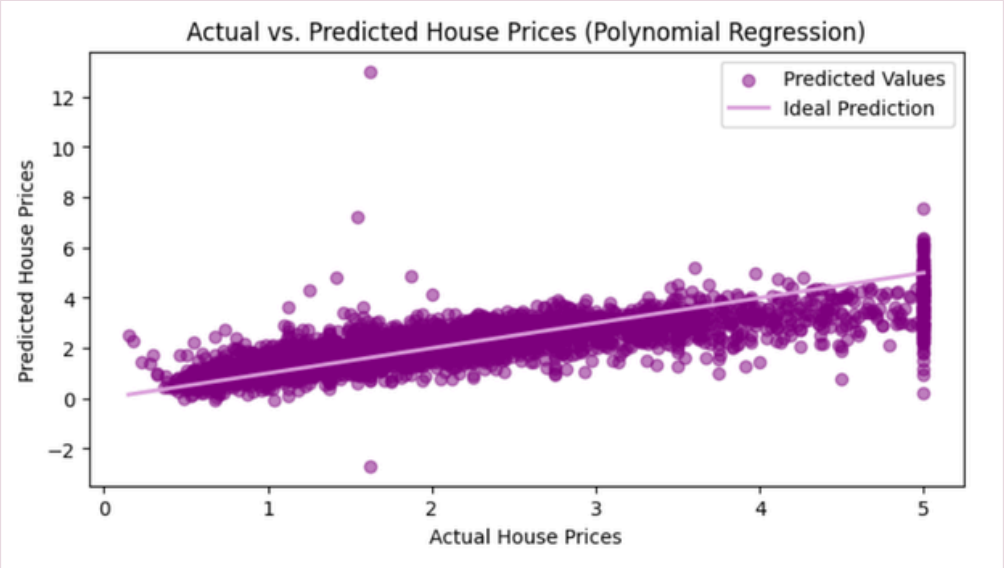

Figure 3.1: Actual vs. Predicted House Prices

Important Features in Predicting House Prices

In predicting house prices, the features that appeared to be the most important were **longitude**, **latitude**, and **MedInc**. This determination was based on visualizing the relationships between these features and house prices using scatter plots. The scatter plot revealed that the relationship between MedInc (Median Income) and house prices was the most pronounced, indicating that house prices tend to increase with rising income levels, as shown in Figure 4.1.

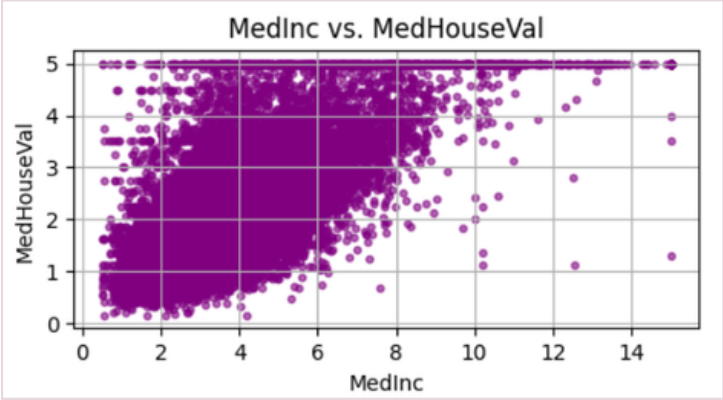

Figure 4.1: MedInc vs. MedHouseVal

Both **latitude** and **longitude** displayed similar clustering patterns, with higher house prices concentrated in specific geographic areas, as illustrated in Figures 4.2 and 4.3. These clusters may be attributed to geographic factors such as proximity to the coast or major urban centers.

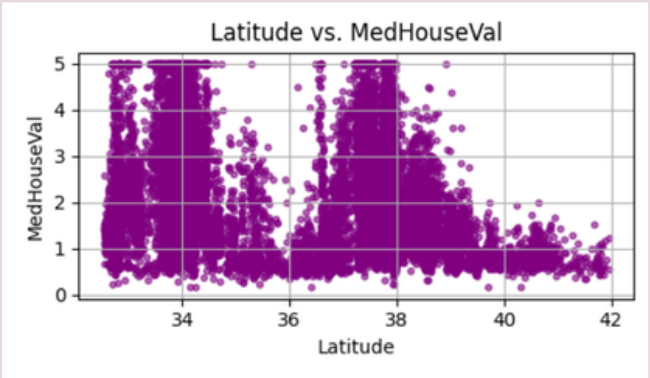

Figure 4.2: Latitude vs. MedHouseVal

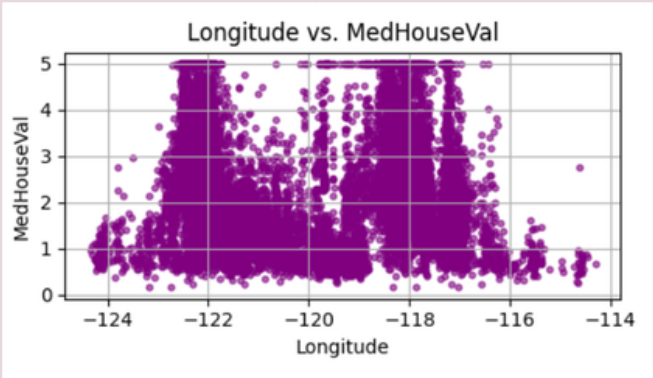

Figure 4.3: Longitude vs. MedHouseVal

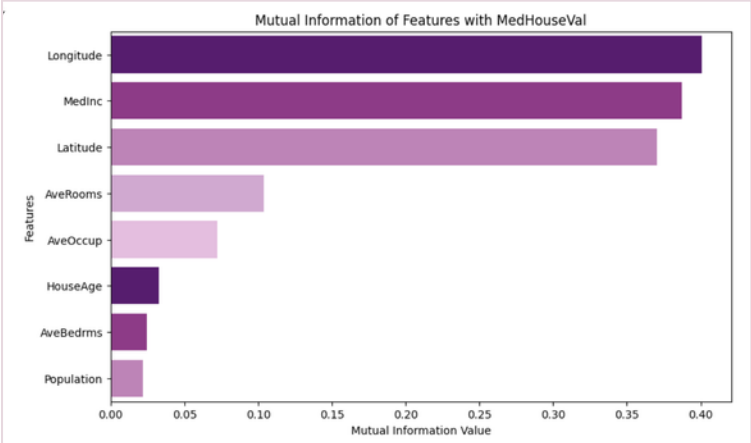

Figure 4.4: Mutual Information Scores for Feature

To gain a clearer understanding of the relationships between the features and house prices, we calculated Mutual Information scores, which quantify how much one variable reveals about another. We chose Mutual Information because it captures the strength of relationships between variables, regardless of whether they are linear or not. The results confirmed the strong relationships between the variables mentioned earlier, as shown in Figure 4.4. This analysis highlights the importance of geographic and economic factors in predicting house prices.

Logistic Regression Model Performance in Classifying Expensive Houses

The logistic regression model performed well in classifying expensive houses, achieving an accuracy score of 0.83 and a ROC-AUC score of 0.91, indicating effective discrimination between classes. Additionally, the model exhibited a precision, recall, and F1-score of 0.82 for the positive class, reflecting its overall strong performance in identifying expensive houses.

Several factors contributed to the model's strong performance:

1. **Significant Features:** The inclusion of relevant features, such as MedInc (Median Income), which has a strong correlation with house prices, allows the model to make accurate predictions.
2. **Effective Class Separation:** The model effectively differentiates between the positive class (expensive houses) and the negative class (not expensive houses), as indicated by the high ROC-AUC score.
3. **Standardization:** Standardizing the features improved model performance by ensuring that all features contribute equally to the distance calculations in the logistic regression algorithm. This can help the model converge faster and achieve better accuracy.
4. **Well-structured Data:** The dataset used for training is relatively clean, with no missing values, which enhances the model's ability to learn from the data.

Linear Regression vs Polynomial Regression

The linear regression model perform poorly due to its inability to capture the complex, non-linear relationships between house prices and features. It assumes a simple straight-line relationship, leading to a Mean Squared Error (MSE) of 0.56 and an R-squared score of 0.58, explaining only 58% of the variance in house prices.

In contrast, the polynomial regression model (degree 2) performed better, with an MSE of 0.46 and an R-squared scoreof 0.65, explaining 65% of the variance. This improvement highlights the polynomial model’s ability to capture more complex patterns, as seen in the scatter plots of features like Latitude and longitude.

Why Polynomial Regression Performs Better:

1. **Capturing Non-linearity:** Housing prices are affected by a variety of factors that often interact in complex, nonlinear ways. For example, Housing prices may rise with increasing income but remain stable when income is higher, or fluctuate based on geographic factors such as latitude and longitude. Polynomial regression can capture these nuances more effectively than linear models.
2. **Feature Interactions:** Features like Latitude and Longitude might have a combined effect on prices. Polynomial regression allows the model to account for interactions between features, which can be crucial in capturing the effect of geographical location.
3. **Sensitivity to Local Trends:** Polynomial regression can effectively detect local trends in the data, capturing variations in house prices that are specific to certain areas or conditions. This capability allows the model to accurately represent the complexities of the California housing dataset and respond to price changes based on local factors.

Limitations and Improvement Strategies

Limitations of the Models Used

1. **Logistic Regression Limitations:**
   - **Linearity Assumption:** Assumes a linear relationship between independent variables and the log-odds of the target variable, which may not capture complex patterns in the data.
   - **Sensitivity to Outliers:** Performance can be significantly affected by outliers in the dataset, potentially skewing predictions.
2. **Polynomial Regression Limitations:**
   - **Overfitting Risk:** Higher-degree polynomials can lead to overfitting, capturing noise rather than the underlying trend.
   - **Complexity and Interpretability:** Models may become complex, making it difficult to interpret the relationships between variables.
3. **Data Limitations:**
   - **Data Clustering:** Certain variables, such as **AveRooms**, **AveBedrms**, **Population**, and **AveOccup**, exhibited clustering around specific values, as shown in Figures 4.5, 4.6, 4.7, and 4.8. This clustering may introduce bias and limit the model's generalizability. Such clustering could negatively impact the model's performance, making it challenging to identify broader patterns that extend beyond those specific values.

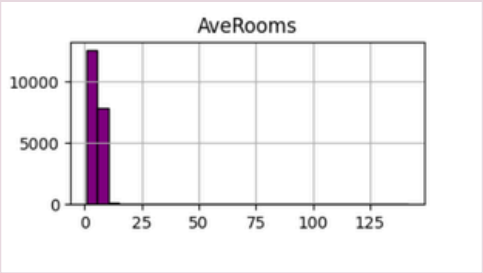

Figure 4.5: Distribution of AveRooms

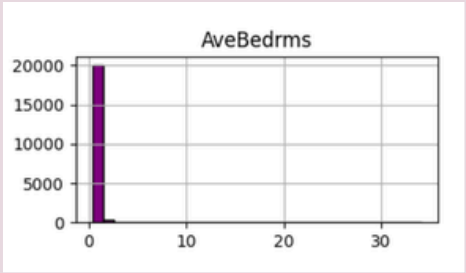

Figure 4.6: Distribution of AveBedrms

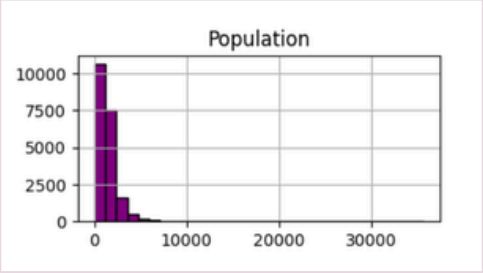

Figure 4.7: Distribution of Population

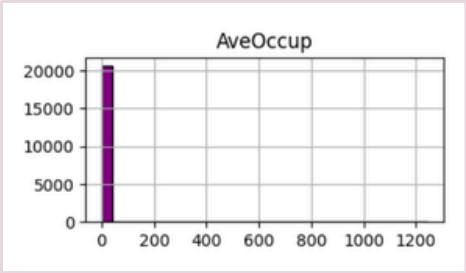

Figure 4.8: Distribution of AveOccup

Recommendations for Improvement

1. **Cross-Validation:** Implement k-fold cross-validation to ensure that the model's performance is consistent across different subsets of the data, helping to identify overfitting or underfitting.
2. **Hyperparameter Tuning:** Optimize hyperparameters for polynomial regression, such as the degree of the polynomial, to find a balance between underfitting and overfitting.
3. **Data Preprocessing Techniques**:
   - **Feature Scaling**: Use methods like StandardScaler or MinMaxScaler to reduce the impact of clustering on the model.
   - **Data Transformation**: Apply transformations such as logarithmic or Box-Cox to minimize clustering effects and improve variable distribution.
   - **Outlier Treatment:** Investigate and potentially remove outliers or use robust regression techniques that are less sensitive to them